ADDITIVE MANUFACTURING · INDUSTRIES

Risk for human activities and structures

Risk per person and per hour, four classes of parts across aviation, cars, trains and medical devices, and six failures of Class A structures.

Draft · November 2026



PART 1

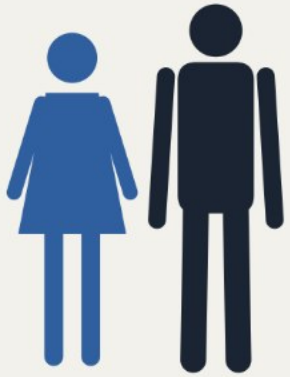
Risk in a human life

How many hours a life lasts, and how much risk each hour of an activity carries.

- Hours in a life
- Risk per person and per hour
- Sports, medical, car, train, aviation

RISK IN A HUMAN LIFE

80 years of life ≈ 701 000 hours



80 years × 365.25 days × 24 h ≈ **701 000 hours** ≈ 7×10^5 h



Work 10 %, travel 5 %, study 2 %: indicative shares of a European life

Check with everyday life. The average risk of dying, all causes, is about 24 micromorts a day, that is **1×10^{-6} per hour**. Summed over a life: 7×10^5 h × 10^{-6} /h ≈ **0.7**, of order one — the hourly scale is consistent with one life.

Sports

Skis, bindings, helmets, harnesses

per hour

10^{-7} – 10^{-5}

hours in a life

ski 1 500 h · scuba 600 h

lifetime risk

ski 1 in 5 000
scuba 1 in 250

Medical

Stents, hip and knee prostheses

per hour

≈ 10^{-6} device events

hours in a life

130 000 h (24 h/day, 15 y)

lifetime risk

≈ 1 in 10 events, not deaths

Car

Car occupant

per hour

$1 – 3 \times 10^{-7}$

hours in a life

23 000 h (1 h/day, 62 y)

lifetime risk

1 in 150 – 1 in 350

Train

Rail passenger

per hour

$2 – 3 \times 10^{-8}$

hours in a life

8 000 h (100 h/y, 80 y)

lifetime risk

≈ 1 in 5 000

Aviation

Airline passenger

per hour

$3 \times 10^{-8} – 4 \times 10^{-7}$

hours in a life

1 200 h (20 h/y, 60 y)

lifetime risk

1 in 2 000 – 1 in 30 000

Space

Launcher, capsule, heat shield

per hour

≈ 10^{-2} at ascent

hours in a life

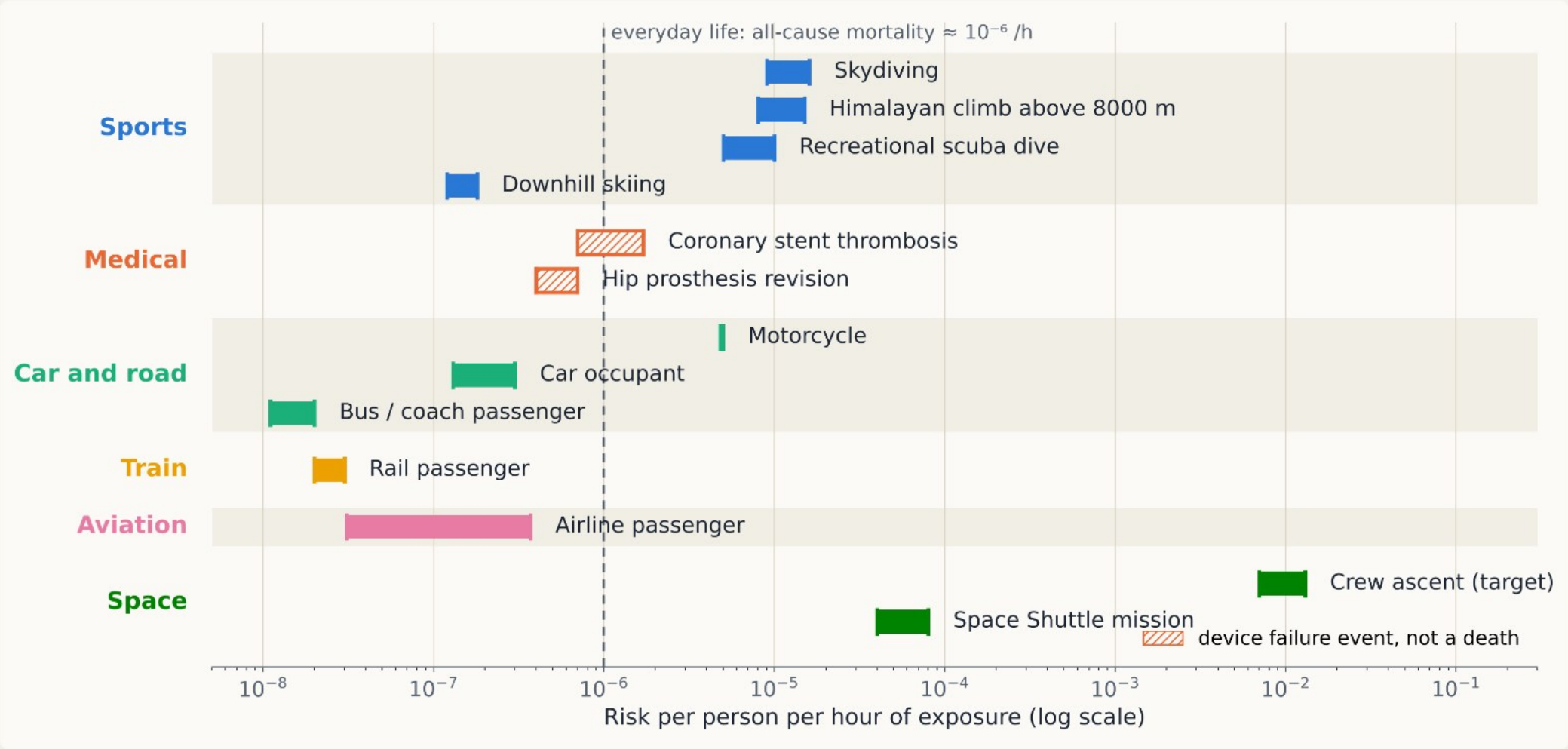
one mission ≈ 250 h

lifetime risk

1 in 68 per Shuttle flight

MEASURING RISK

Risk per person and per hour, by category



Sources: ETSC (EU), UK DfT, BSAC, USPA, NASA, ABSORB III, joint registries · data: 02_technologies/tables/02_risk_per_hour.csv

MEASURING RISK

How the numbers are estimated, and rules of thumb

Estimating a rate

1. **Count the events**
   Deaths, or device failures, over a period: accident statistics, clinical registries.

2. **Divide by the exposure**
   Persons × hours: travel surveys, flight hours, ski days, implant-years × 8 766 h.

3. **Convert and bracket**
   Per km or per trip → per hour with the average speed or duration; keep a range, not one number.

1 micromort (10^{-6}) ≈ 370–600 km by car, 1 600 km by jet, 10 000 km by train

Lifetime = rate × hours: 1 h a day for 60 years ≈ 2×10^{4} h

Class A design target 10^{-9} per flight hour: 1 000 × below everyday life

Orders of magnitude per hour

10^{-8} /h	Train, bus
10^{-7} /h	Car, airliner, skiing
10^{-6} /h	Everyday life, all causes
10^{-5} /h	Motorbike, scuba, high mountains
10^{-2} /h	Launch to orbit

Micromort distances: Spiegelhalter, UK data (Wikipedia, Micromort); rates as on the previous slide

PART 2

Four classes of risk

The consequence of a failure sets the class; the class sets the allowed probability and the effort.

- **A** · catastrophic or hazardous
- **B** · major
- **C** · minor
- **D** · no safety effect

AVIATION · EASA CM-S-008 · CS 25.1309

Four classes of parts, set by the effect of their failure

Class	Effect of the part's failure	Condition	Allowed per flight hour	Typical parts
A · High	Directly affects continued safe flight and landing; serious or fatal injuries	Catastrophic / Hazardous	$< 10^{-9}$ / $< 10^{-7}$	Engine discs, fan blades, wing spars, pressure bulkhead, flight-control actuators
B · Medium	Indirectly affects safe flight; minor injuries, large rise in crew workload	Major	$< 10^{-5}$	Secondary structure, fuel and hydraulic lines, brakes
C · Low	No effect on safe flight; slight loss of function or comfort	Minor	$< 10^{-3}$	Cabin interior: bin latches, seat parts, ducts
D · Negligible	No safety effect	No safety effect	No requirement	Trim, covers, placard holders

Two rules for Class A: below 10^{-9} per flight hour, and no single failure may lead to a catastrophe, however unlikely.

Targets apply to a failure condition of the system, not to one part. Structure relies on safety factors, A/B-basis allowables and damage tolerance.

EASA CM-S-008 Issue 04 (2025), Table 1, from ASTM F3572-22; CS-25 / AMC 25.1309; FAA AC 25.1309-1B

CARS · ISO 26262

Safety integrity levels for electronic functions

S **Severity**
S0 no injury … S3 fatal

E **Exposure**
How often the situation occurs, E0 … E4

C **Controllability**
Can the driver avoid harm? C0 … C3

≈ class	ASIL	Random hardware failure target	Typical systems
A	D	$< 10^{-8}$ /h	Electric power steering, brake-by-wire, ESC/ABS, airbag
A / B	C	$< 10^{-7}$ /h	Adaptive cruise control, torque control
B	B	$< 10^{-7}$ /h	Headlights, brake lights, speedometer
C	A	none (~10^{-6})	Rear lights, driver-assist warnings
D	QM	quality management	Infotainment, comfort

ISO 26262 covers electrical and electronic systems only. Wheels, brake discs and suspension arms go through UNECE type approval, FMEA and safety-critical part marking.

Ten times looser than aircraft: the driver can often recover, a crash involves few people — but 1.5 billion cars turn 10^{-8} into real events. Worst recalls were mechanical: **GM ignition switch (2014), Takata inflators.**

TRAINS · EN 50126 / 50128 / 50129

Safety integrity levels for signalling, inspection for steel

≈ class	SIL	Tolerable hazard rate	Typical functions
A	4	10^{-9} – 10^{-8} /h	Interlocking, train protection (ETCS on-board)
B	3	10^{-8} – 10^{-7} /h	Safety functions with a fallback (project-specific)
C	2	10^{-7} – 10^{-6} /h	Level-crossing and door functions, by line
C / D	1	10^{-6} – 10^{-5} /h	Monitoring and diagnostic aids

Steel is not in the SIL table

- SILs apply to signalling and control electronics
- Wheels (EN 13979-1), axles (EN 13103/13104) and rails: designed for fatigue, kept safe by inspection
- One wheel, one rail weld: a single load path, a Class A part
- Eschede and Adamuz, in Part 3, are both mechanical

$\approx 2 \times 10^{-8}$ /h

rail passenger risk, the lowest of all modes: about 1 death per 30 billion passenger-km in the EU

IEC 61508; EN 50126, EN 50128, EN 50129; ERA Safety Overview 2021

MEDICAL DEVICES · EU MDR 2017/745

Stents and prostheses: classed by invasiveness

≈ class	MDR class	Examples
A	III	Coronary stents; hip, knee, shoulder replacements; spinal cages
B	IIb	Other implants: bone plates, screws, pedicle screws and rods
C	IIa	Short-term invasive devices, airway stents (by use)
D	I	External limb prostheses, orthoses (self-certified)

No probability per hour in the rules. ISO 14971 asks to reduce risk as far as possible, then weigh it against the clinical benefit: the patient is already at risk without the device.

Failures appear years later, in clinical registries, not in the lab.

≈ 5 %

of hip replacements revised within 10 years · ≈ 6×10^{-7} per hour of use

0.7 %

stent thrombosis in the first year (Xience) · ≈ 8×10^{-7} per hour

1.5 %

with the bioresorbable Absorb scaffold: withdrawn in 2017 after 150 000 patients

EU MDR Annex VIII, Rule 8; ISO 14971; Ellis et al., ABSORB III, JACC 2017; national joint registries

SPACE · CREWED, UNCREWED, PASSENGERS

Space: risk accepted per mission, not designed per hour

Crewed vehicles

1 in 270

loss of crew per ISS mission (NASA Commercial Crew requirement)

- Ascent ≤ 1 in 500, entry ≤ 1 in 500
- On orbit, micrometeoroids and debris drive the risk
- Actual: Shuttle 2 losses in 135 flights; Soyuz 2 fatal flights (1967, 1971)

Uncrewed vehicles

≈ 96 %

of orbital launches reached orbit in 2025 (about 317 of 325); first flights fail far more often

- Payload classes A–D (NASA NPR 8705.4A): A very low risk (JWST) … D high risk (CubeSats)
- A class A payload flies on a launcher with a long record
- A lost probe costs money and years, not lives

Passengers and space travel

≈ 10^{-2} /h

during ascent: about 10^{5} × an airliner hour

- US: the vehicle is licensed for the safety of the public on the ground
- Spaceflight participants sign an informed consent (14 CFR 460.45): no occupant safety level is certified

Read with Part 3: the Shuttle was flown with a claimed 1 in 100 000 and an observed 1 in 68.

ONE SCALE

Four classes across domains

Class	Aviation (per flight hour)	Rail (SIL)	Car (ASIL)	Medical (EU MDR)	Space
A	Catastrophic < 10^{-9}, hazardous < 10^{-7}	SIL 4: 10^{-9} – 10^{-8} /h	ASIL D: < 10^{-8} /h	III: stents, joint replacements	Crew: 1 in 270 per mission; payload class A (JWST)
B	Major < 10^{-5}	SIL 3: 10^{-8} – 10^{-7} /h	ASIL C / B: < 10^{-7} /h	IIb: plates, screws	Payload class B (Mars Reconnaissance Orbiter)
C	Minor < 10^{-3}	SIL 2 / 1: 10^{-7} – 10^{-5} /h	ASIL A	IIa	Payload class C
D	No safety effect	not safety-related	QM	I: external prostheses	Payload class D (CubeSats)

Same logic everywhere: the consequence of failure sets the class, the class sets the target and the effort: redundancy, testing, inspection. **The mapping between domains is indicative.**

NASA Commercial Crew loss-of-crew requirements; NASA NPR 8705.4A payload risk classes

PART 3

Catastrophes with and without a Class A failure

Four Class A parts that failed; three catastrophes started by a lower-class part or by management.

- A Class A part fails: Sioux City, Eschede, Ariane 5 ECA, Adamuz
- Lower class or management: Challenger, Columbia, Qantas 32
- What inspection could not see
- Conclusion

CASE STUDIES

Seven failures on one time line, two routes

[Picture: booster plume] *nasa.gov*

1986

Challenger

Known joint erosion, constraint waived

1989

Sioux City

Titanium fan disc with a melt defect



1998

Eschede

Fatigue crack inside a wheel tyre



[Picture: Vulcain 2 nozzle] *esa.int*

2002

Ariane 5 ECA

Vulcain 2 nozzle cooling tubes crack

2003

Columbia

Foam strike breaches the wing

[Picture: foam impact test] *nasa.gov*



2010

Qantas 32

Unclassified oil pipe, turbine disc burst

2026

Adamuz

Rail broken at a weld

[Picture: rail fracture] *CIAF note*

A Class A part fails by itself

Sioux City · Eschede · Ariane 5 ECA · Adamuz

A lower-class part or management starts it

Challenger · Columbia · Qantas 32

Contrasts

Gelida 2026: a motorway wall falls on a train · Santiago 2013: overspeed, no part failed

CLASS A FAILURE · AVIATION · 1989

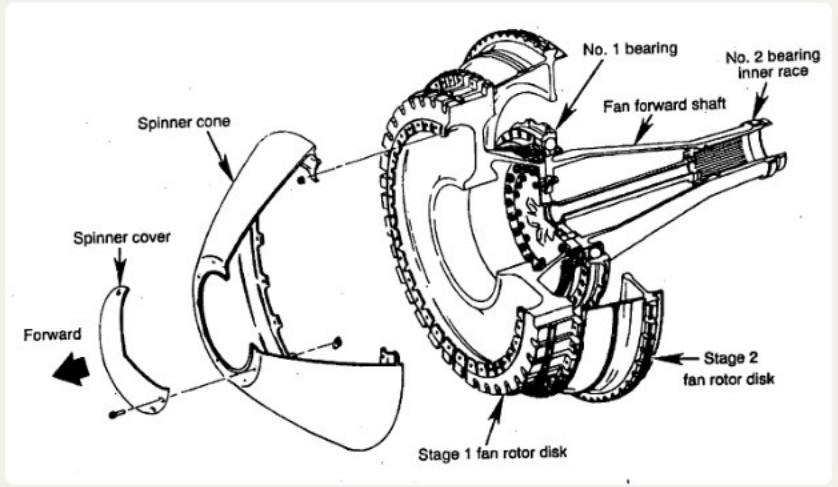
Sioux City: one fan disc, three hydraulic systems lost

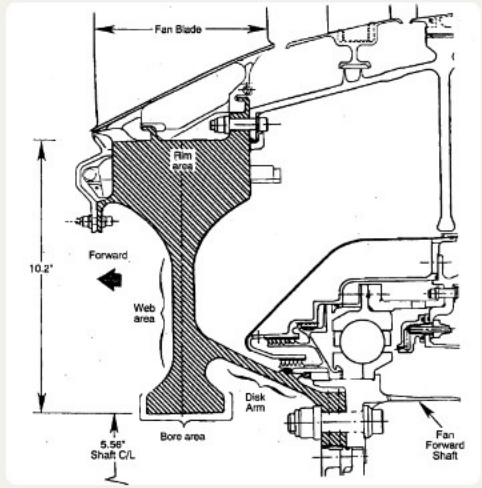
111

deaths among 296 on board; 185 survived

- 19 July 1989, United 232, cruise at 11 300 m
- The 168 kg titanium stage 1 fan disc of the tail engine bursts
- Debris cuts the lines of all three hydraulic systems: no flight controls
- The crew steers with engine thrust alone to Sioux City

Lesson: three separate hydraulic systems, routed close together: one high-energy event defeated the redundancy.





CF6-6 fan rotor and stage 1 fan disc section; disc height 259 mm, bore radius 141 mm (NTSB AAR-90/06, fig. 5, 17)

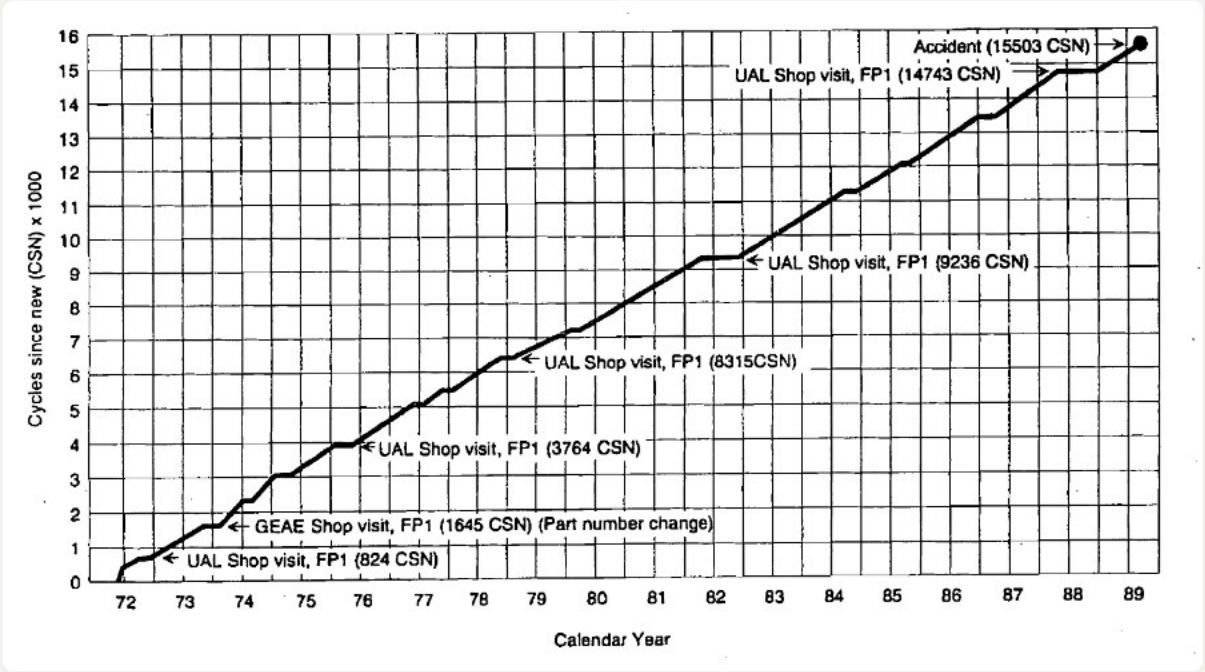


Reconstructed No. 2 fan disc (FAA Lessons Learned)

Found 3 months later in farm fields near Alta, Iowa: two pieces making up nearly the whole disc.

CLASS A FAILURE · AVIATION · 1989

Sioux City: a melt defect, six inspections, one missed crack



Cycles since new and the six fluorescent penetrant inspections (NTSB AAR-90/06, fig. 9)

- 1971: a nitrogen-rich hard alpha inclusion forms while melting the titanium (double vacuum melt)
- A 1.4 × 0.4 mm cavity in the bore starts a fatigue crack, 31.5 mm long and 14.2 mm deep at failure
- Six inspections accepted the disc, the last one 760 flights before the accident

15 503 cycles

at failure, against a life limit of 18 000

Why it was missed: a defect born at the melt shop, a safe-life rule that assumes defect-free material, and a manual inspection that depends on the inspector's eye. **Safe life alone was not enough.**

CLASS A FAILURE · AVIATION · AFTER 1989

Sioux City: from safe life to probabilistic damage tolerance

1989
Ultrasonic inspection of CF6-6 fan discs (AD 89-20-01)

1990–91
FAA titanium review team: triple vacuum melting, damage tolerance

1995
FAA–SwRI Turbine Rotor Material Design programme with four engine makers

1997
DARWIN code, version 1: probability of fracture from hard alpha defects

2001
AC 33.14-1: damage tolerance for titanium rotors, target risk per flight cycle

2007
14 CFR 33.70: damage tolerance required for engine life-limited parts

2009
AC 33.70-1 and AC 33.70-2 (hole features)

2023
AC 33.70-3, titanium anomalies, replaces AC 33.14-1

DARWIN (Design Assessment of Reliability With INspection): finite elements + fracture mechanics + inspection simulation + Monte Carlo, giving the probability of fracture of a disc with rare defects. R&D 100 award, 2000.

Airframe and Europe: hydraulic fuses and shut-off valves on the DC-10 (AD 90-13-07, 95-14-06); AC 20-128A on uncontained rotor debris; EASA CS-E 515, engine critical parts.

FAA Lessons Learned N1819U; SwRI DARWIN background; 72 FR 50856 (2007); FAA AC 33.14-1, 33.70-1/-2/-3

CLASS A FAILURE · RAIL · 1998

Eschede: a wheel tyre breaks at 200 km/h

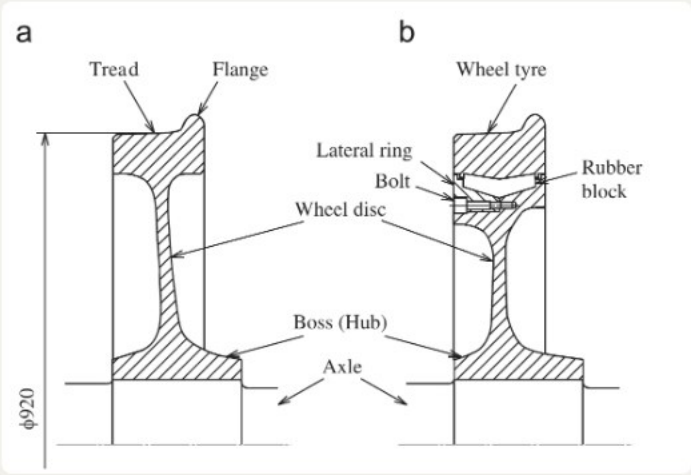
101

deaths, 88 seriously injured: the world's worst high-speed rail accident

- 3 June 1998, ICE 884 Munich → Hamburg
- The steel tyre of a rubber-sprung wheel breaks by fatigue, peels off and pierces the floor
- 6 km later it catches a set of points: coaches derail and hit a road bridge, which collapses on the train



The accident site (Richard et al. 2005, fig. 1)



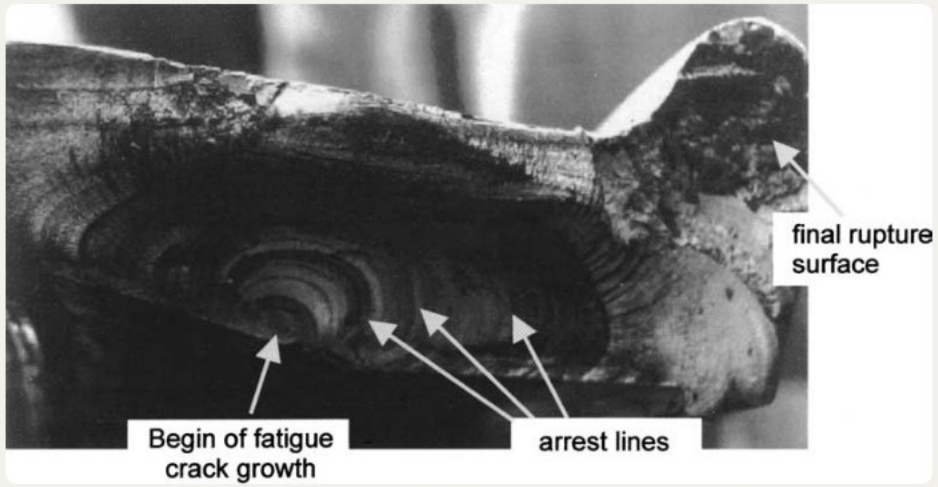
(a) monobloc, (b) rubber-sprung wheel, Ø 920 mm (Kubota & Hirakawa 2009)

The rubber-sprung wheel BA 064

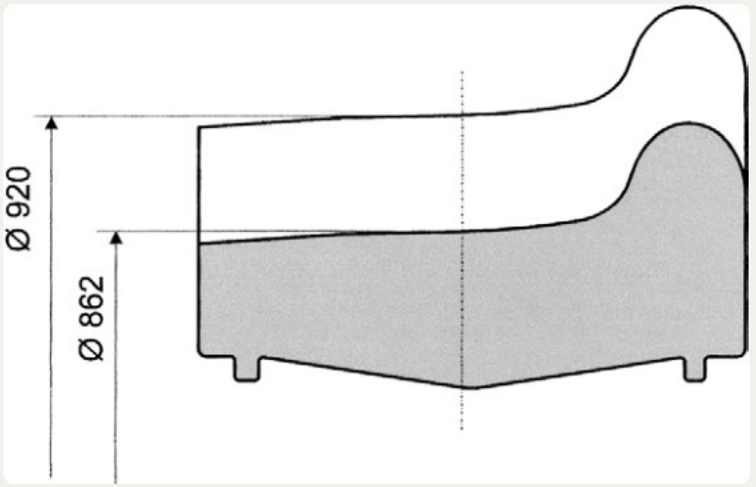
- Steel tyre on 34 pre-compressed rubber blocks, bolted to the wheel disc by a lateral ring
- Used for decades on trams and metros; admitted on ICE 1 trailer cars in autumn 1992 to cut vibration and noise

CLASS A FAILURE · RAIL · 1998

Eschede: a fatigue crack inside a worn tyre



Fracture surface: origin on the inner face, arrest lines (Richard et al. 2005)



Wear and re-profiling: Ø 920 → 862 mm (Richard et al. 2005)

60 → 31 mm

tyre thickness by wear; scrap limit Ø 854 mm

> 3.5

fatigue safety factor at Ø 862 mm, 80 kN

Fracture mechanics (Richard 2005)

- Highest hoop stress on the hidden inner face
- A 1.5 mm crack grows slowly, with arrests, long before rupture

Full-scale tests (Kubota 2009)

- No fretting damage from the rubber contact
- Test crack starts at the centre of the inner face, as in service

Lesson: the design margin looked sufficient; the missing barrier was an inspection able to see the inner face. **ICE 1 went back to monobloc wheels.**

CLASS A FAILURE · SPACE, UNCREWED · 2002

Ariane 5 ECA flight 157: the Vulcain 2 nozzle gives way

- 11 December 2002, Kourou: first flight of the 10-tonne Ariane 5 ECA, two satellites on board, no crew
- A leak opens in the cooling circuit of the Vulcain 2 nozzle extension; the nozzle overheats and loses its integrity
- Asymmetric thrust: control of the trajectory is lost, about 3 minutes after lift-off; the launcher is destroyed

[Picture: Vulcain 2 engine and nozzle extension]
esa.int › Ariane 5 ECA; ariane.group

Most probable cause (Inquiry Board, January 2003)

- Fissures in the cooling tubes that form the nozzle structure: degraded thermal state
- Flight loads on the Vulcain 2 not defined in full, and hard to reproduce in ground firings
- Differences from Vulcain 1 (12 successful flights): the shape of the cooling tubes and the stiffener technology

What changed

- Nozzle redesigned with the Vulcain 1 experience
- Ground simulation of the loads seen in flight
- Better quality of flight equipment; ECA back in 2005
- Rosetta, due on an Ariane 5 in January 2003, flew in 2004 to another comet

Lesson: a single nozzle with no backup is a Class A part even with no one on board; it was qualified on the ground for loads that only flight produced. **Define the loads before the class can be trusted.**

Source: Arianespace / ESA, Flight 157 Inquiry Board findings, 7 January 2003

CLASS A FAILURE · RAIL · JANUARY 2026

Spain: two accidents in three days

Adamuz (Córdoba), 18 January · high-speed line

1. **Rail broken at a weld before the train passed**
   Notches on the wheels of three earlier trains match the break
2. **Iryo 6189 Málaga → Madrid: rear coaches derail**
   The rail tips outward under coach 5; coach 6 leaves the track
3. **Alvia 2384 on the other track hits the coaches**
   46 deaths: Spain's worst high-speed accident

Gelida (Barcelona), 20 January

- Storm after months of drought: a motorway retaining wall falls onto the R4 commuter line
- It hits the cab as the train arrives: 1 trainee driver killed, 37 injured
- The whole Rodalies network stops for inspection

Lesson: infrastructure is a Class A structure shared by every train: a hidden rail defect was hit by several trains before one derailed. **Final causes still under investigation (CIAF).**

CIAF preliminary notes, 23 January 2026 (via Railway Gazette); Ara, March 2026; Catalan News

LOWER CLASS OR MANAGEMENT · SPACE · 1986 / 2003

Space Shuttle: known anomalies accepted by management

1 in 100 000
loss of a flight, as quoted by management before Challenger

≈ 1 in 100
Feynman's estimate after Challenger, from engineers' data

2 in 135
missions actually lost: about 1 in 68

Challenger, 28 January 1986

- Booster field-joint O-rings: Criticality 1 since 1982, no backup — a Class A item
- Erosion seen on earlier flights; the launch constraint was waived again and again
- Cold launch decided against the engineers' advice: 7 crew lost

Columbia, 1 February 2003

- Foam insulation, not a structural item, shed at nearly every launch: treated as 'in family'
- A foam block breaches the wing's carbon leading edge, a Class A part
- Hot plasma enters the wing at re-entry: 7 crew lost

Lesson: both chains passed through management decisions on known anomalies. **A NASA review later put the first Shuttle flights at about 1 in 10.**

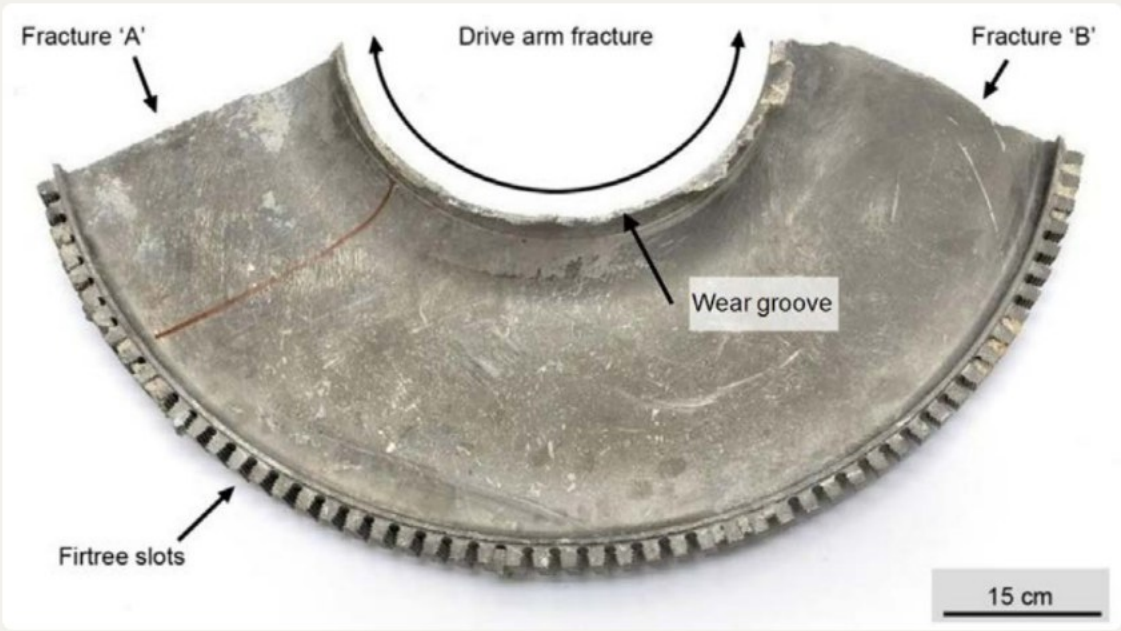
Rogers Commission (1986), ch. V and app. D, F; CAIB report (2003); Hamlin et al., NASA JSC (2011)

LOWER CLASS STARTS IT · AVIATION · 2010

Qantas 32: an A380 turbine disc bursts after take-off



No. 2 engine, Rolls-Royce Trent 900 (ATSB AO-2010-089, cover)



IP turbine disc: 43 % of the disc, about 70 kg, found on Batam Island, Indonesia (ATSB fig. 38)

469
on board, no injuries

2 100 m
climbing out of Singapore

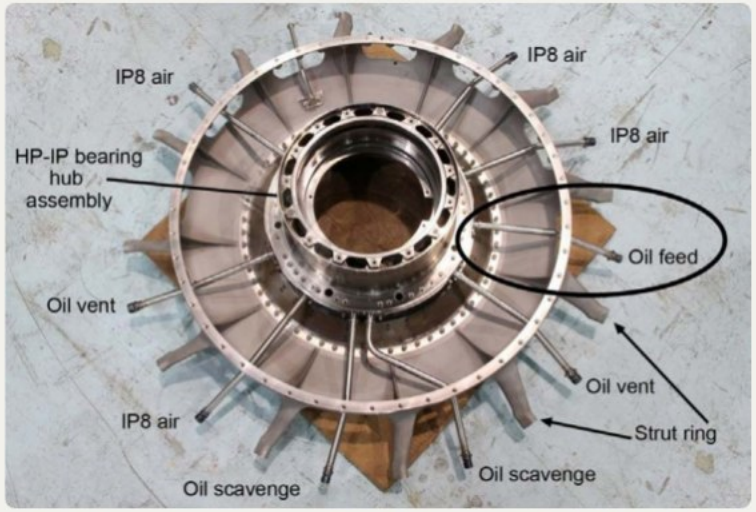
≈ 50 min
to complete the initial ECAM procedures

The disc fragment fell on a property on Batam; other debris pierced the wing and fuel tanks and cut wiring.

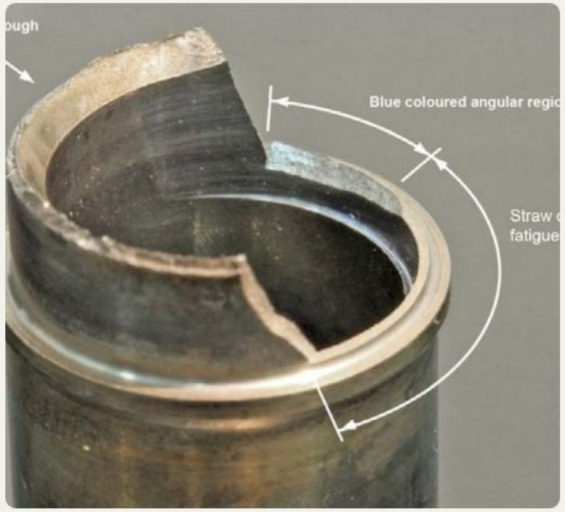
ATSB AO-2010-089 (2013), CC BY 3.0 AU · Source: Australian Transport Safety Bureau

LOWER CLASS STARTS IT · AVIATION · 2010

Qantas 32 root cause: a 0.35 mm wall instead of 0.82 mm



HP/IP bearing support, oil feed circled (ATSB fig. 29)



Fractured oil feed stub pipe (ATSB fig. 43)

0.91 mm
nominal wall in the counter bore

0.82 mm
minimum allowed, worst-case tolerances

0.35 mm
measured at the crack

The bearing hub moves about 6 mm at full thrust and bends the pipe on every flight.

Counter bore machined off-centre during manufacturing

↓

Stub pipe wall 0.35 mm at its thinnest

↓

Fatigue crack, oil leak into a hot buffer space

↓

Internal oil fire: IP turbine disc separates from its shaft

↓

Disc overspeeds and bursts: uncontained debris

LOWER CLASS STARTS IT · AVIATION · 2010

Qantas 32: a Class A disc lost through an unclassified pipe

Part	Manufacturer's class (FMECA)	Effect of its failure	≈ class	Controls that follow
IP turbine disc	Critical part (engine critical, life-limited, CS-E 515)	Hazardous: uncontained high-energy debris	A	Hazardous effects ≤ 10^{-8} per engine flight hour; life limit, special process and inspection controls
HP/IP bearing support with oil feed stub pipe	Unclassified at certification; Reliability sensitive after the event	Oil leak; foreseen in the FMECA to end, extremely rarely, in disc overspeed	D → B	No special controls: the wall thickness was not an explicit drawing feature

Lesson: the failure analysis assumed that every part conforms to its drawing. A low-class part made out of tolerance started a chain that ended at a Class A part. **Classify along the whole failure chain, and match manufacturing control to the class.**

ATSB AO-2010-089, sections 2.4.5 and 2.4.6; JAR-E: extremely remote ≤ 10^{-8} per engine flying hour

WHAT INSPECTION COULD NOT SEE

Details undetectable at inspection

Case	The detail	Why it was not seen	Answer afterwards
Sioux City 1989	Hard alpha inclusion and a 31.5 mm fatigue crack at the disc bore	Six fluorescent penetrant inspections read by eye; the defect started below the surface	Ultrasonic and eddy current, triple melting, damage tolerance
Eschede 1998	Fatigue crack on the inner face of the wheel tyre	Face hidden by the rubber blocks and the wheel disc	Monobloc wheels, ultrasonic inspection
Ariane 5 ECA 2002	Fissures in the nozzle cooling tubes	Flight loads not reproduced by ground firings	Nozzle redesign, ground simulation of flight loads
Adamuz 2026	Rail fracture at a weld	Defect inside the weld; earlier trains marked it without derailing	Under investigation (CIAF)
Qantas 32 2010	Oil feed stub pipe wall of 0.35 mm	Counter bore hidden once the pipe was welded; a CMM error statement was not acted on	Wall thickness checked, hubs replaced, part reclassified
Columbia 2003	Breach in the carbon leading edge of the wing	Not visible from the cabin; no inspection in orbit	Inspection in orbit with a sensor boom and imagery

Rule: a part may only be counted on to be inspected if the defect that kills it can be seen by the method used, where it starts.

NTSB AAR-90/06; Richard et al. 2005; Arianespace 2003; CIAF note 23 Jan 2026; ATSB AO-2010-089; CAIB report 2003

CONCLUSION

Two routes to catastrophe

A Class A part fails

Sioux City · Eschede · Ariane 5 ECA · Adamuz

- A hidden material or manufacturing defect in a single load path
- Calculations were right for the part as drawn, not for the part as made
- Answer: damage tolerance, cleaner material processes, inspection able to see the defect

A lower class or management starts it

Qantas 32 · Columbia · Challenger · Gelida

- An unclassified pipe, foam insulation or an external wall starts the chain
- Known anomalies are accepted flight after flight; constraints are waived
- Answer: classify along the whole failure chain, and never waive a Class A constraint

Estimated risk ≠ real risk: 1 in 100 000 claimed, 1 in 68 observed on the Shuttle.

One event defeats redundancy: debris cut every hydraulic line (Sioux City) and wiring (Qantas 32).

PART 4

References

Standards, risk data, accident reports and papers.

- Classes and targets
- Risk data
- Accident reports and studies

References (1/2)

CLASSES AND TARGETS

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